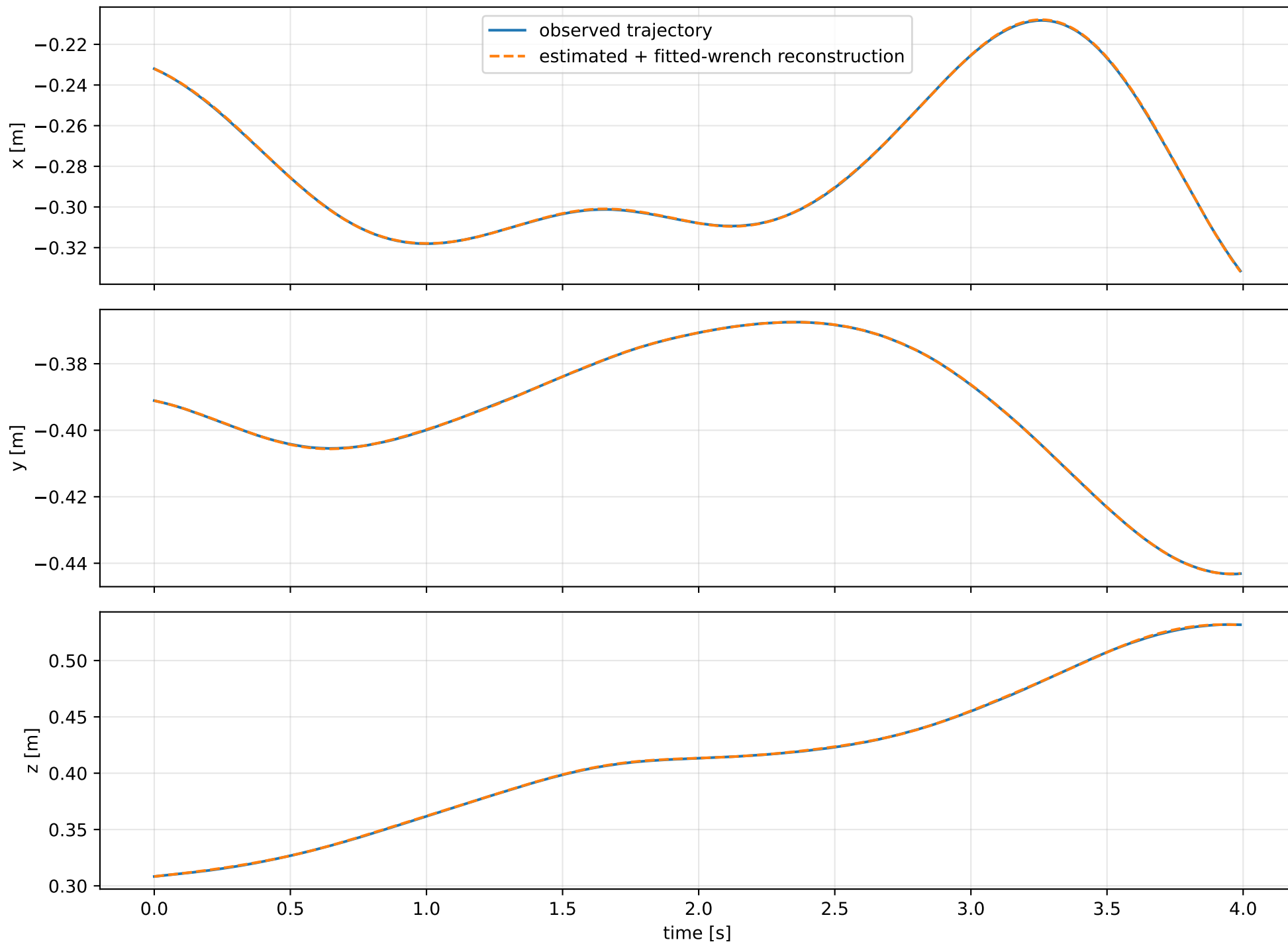
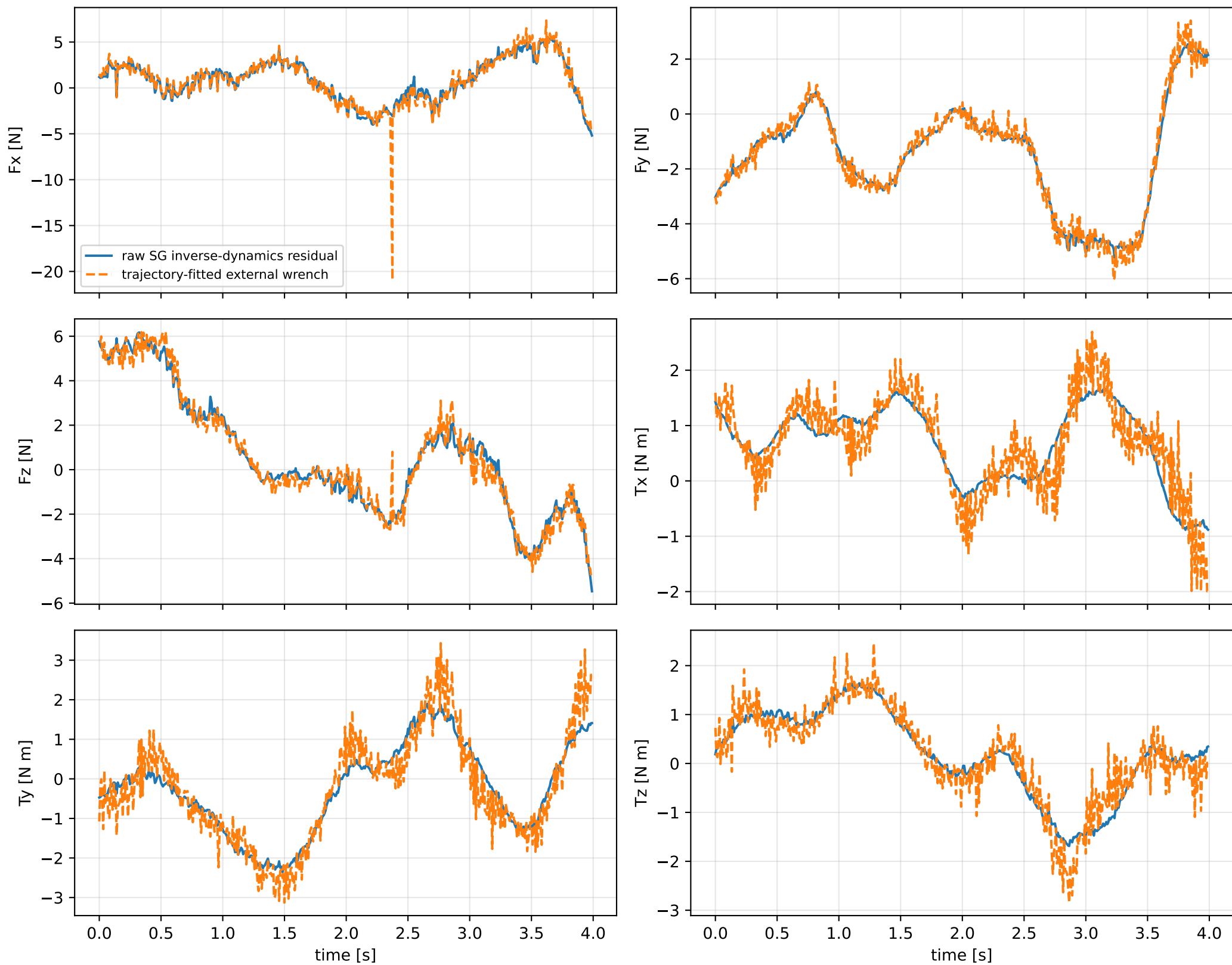


sweep_sg_window_covariance__002: trajectory

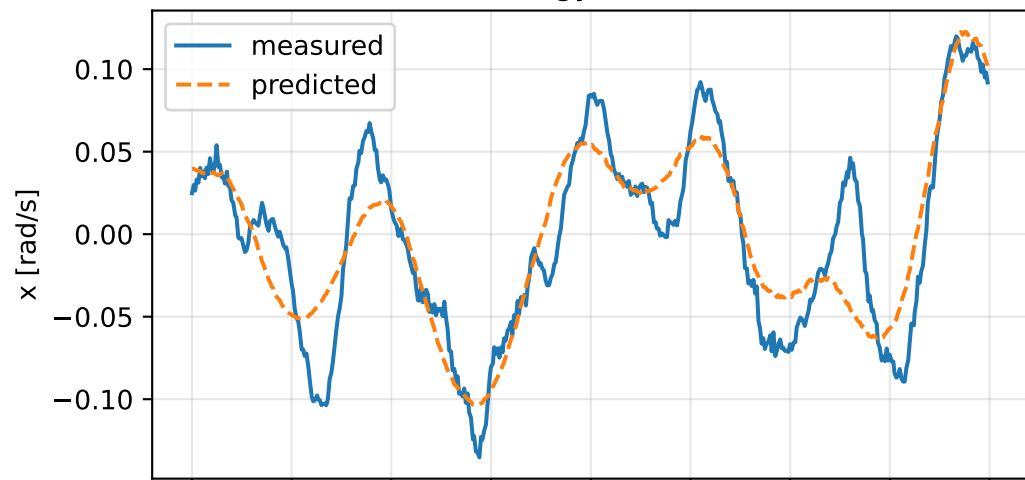


sweep_sg_window_covariance_002: residual wrench

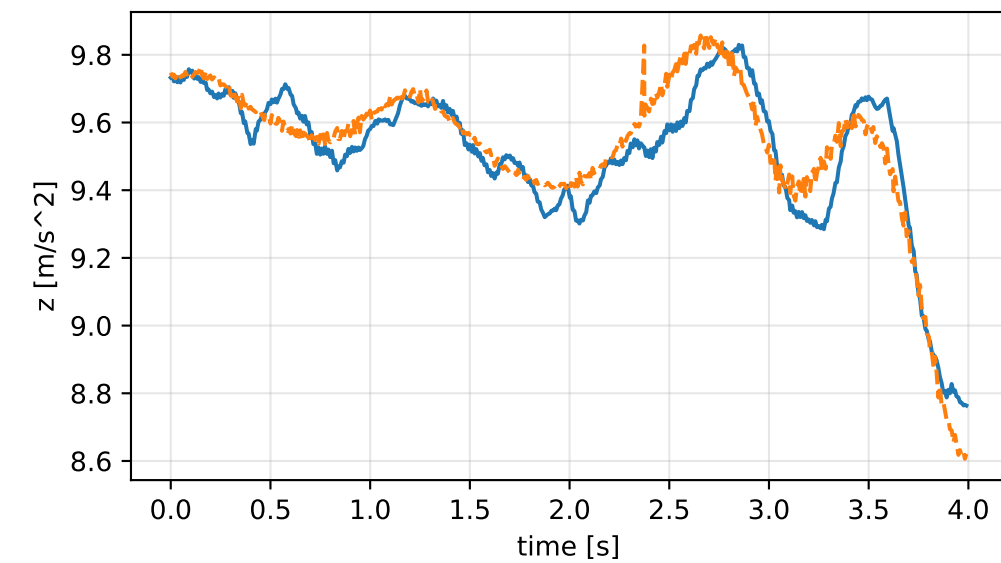
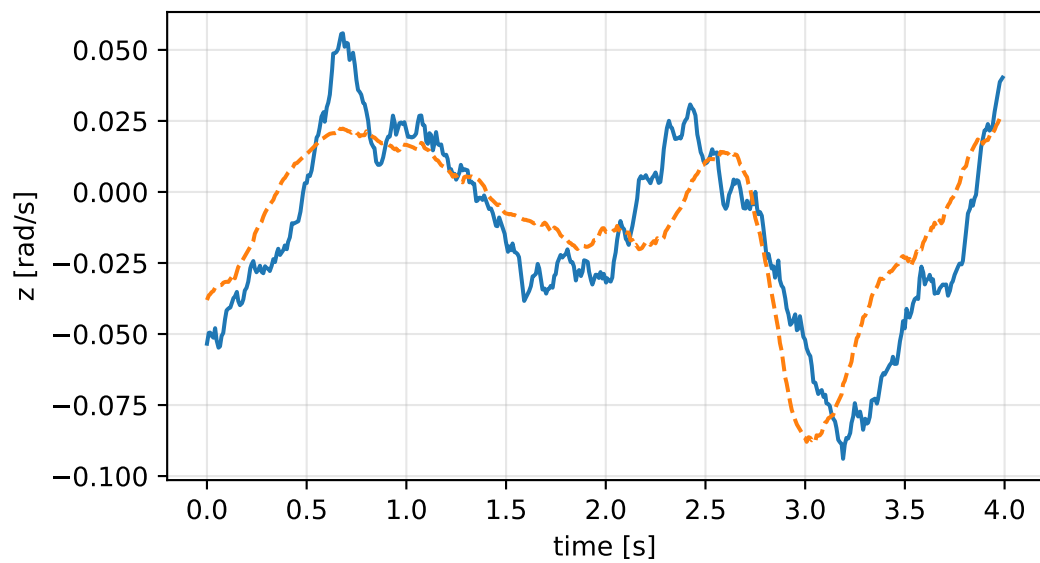
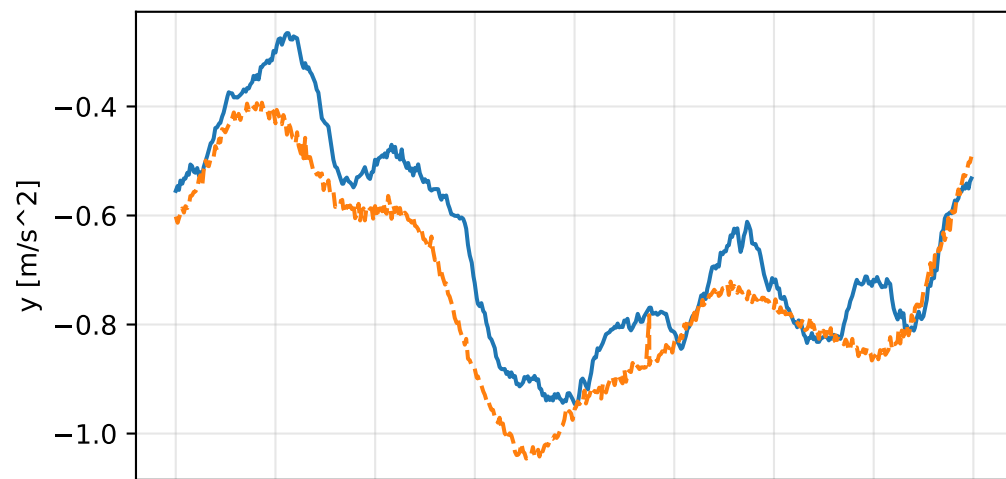
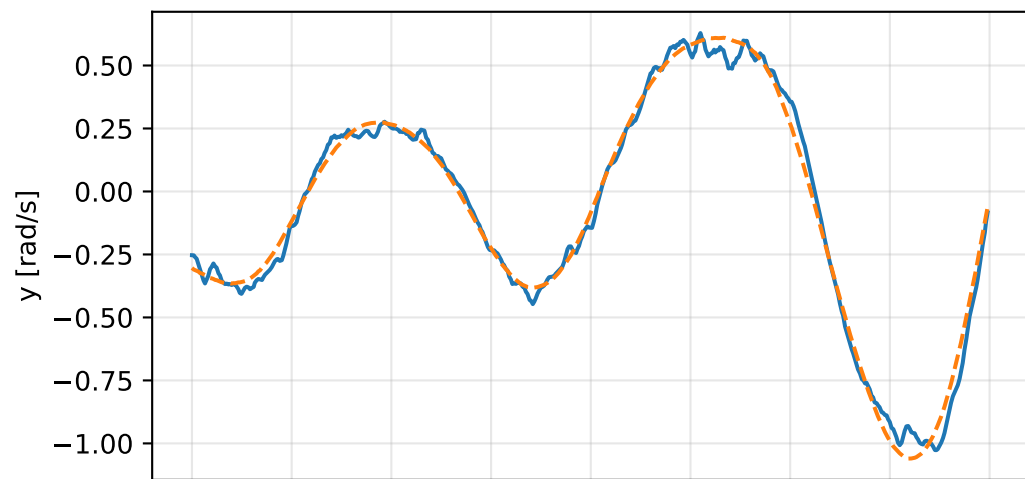
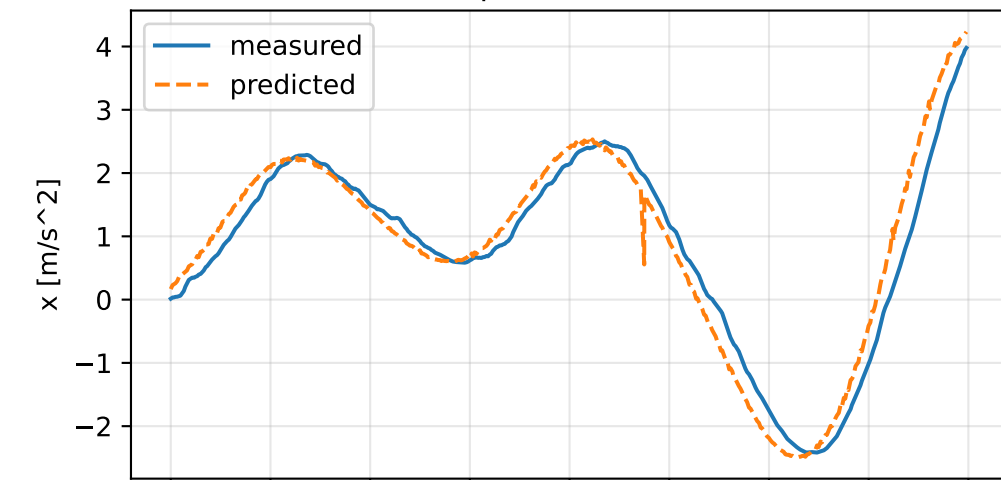


sweep_sg_window_covariance_002: measured sensor comparison

gyro



specific force



sweep_sg_window_covariance__002: nominal -> estimated parameters

Case: sweep_sg_window_covariance__002
status: completed (all stages completed)

mass [kg] nominal 2.35159759 estimated 15.7751224 delta 13.4235248

inertia [kg m^2] (nominal -> estimated; delta)

[6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [2.5309375 -0.5887801 -0.6475513] d=[2.4659375 -0.5887794 -0.6475704]
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [-0.5887801 2.1080627 -1.1357779] d=[-0.5887794 2.04311 -1.1357779]
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [-0.6475513 -1.1357779 3.0165307] d=[-0.6475704 -1.1357779 2.8875386]

CoG [m] [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [-0.124789 0.1085478 -0.0967128] d=[-0.1227643 0.1085784]

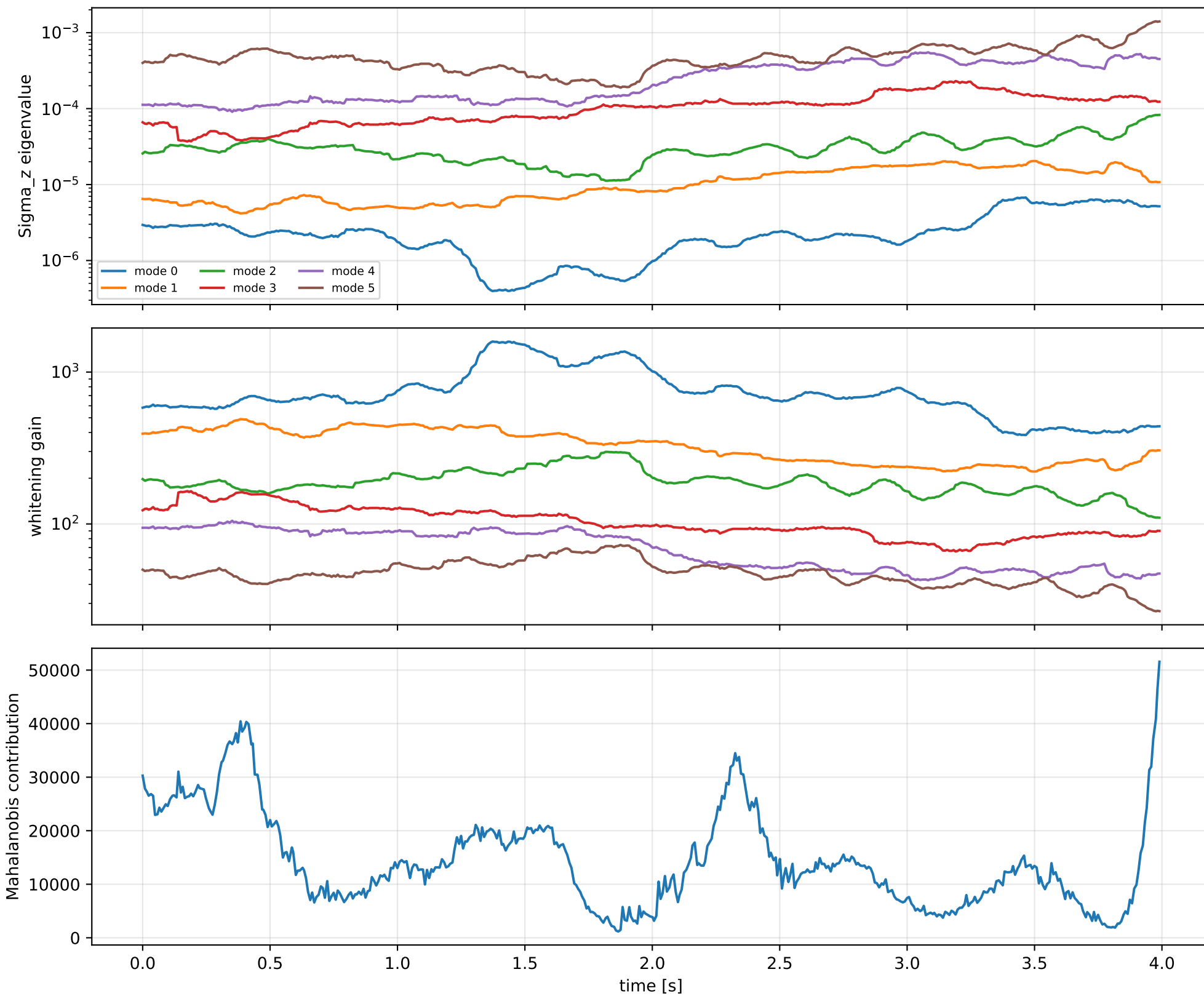
rotor force effectiveness

[1. 1. 1. 1.] -> [6.3172169e+00 5.1572888e-05 5.3482430e+00 1.0613867e+01] d=[5.3172169 -0.9999484 4.348243 9.6138675]

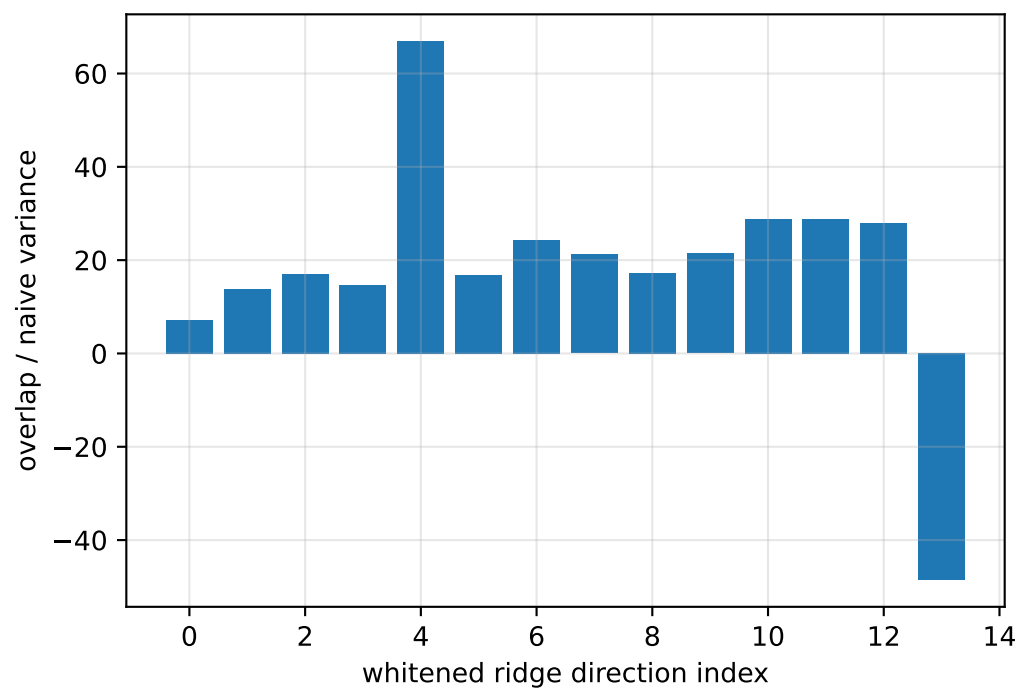
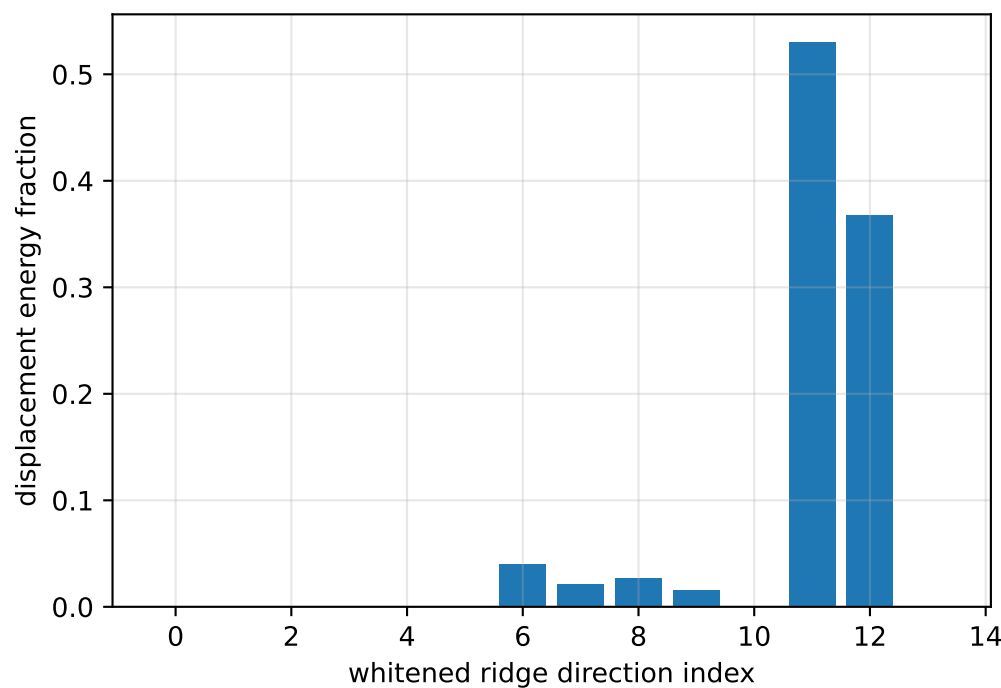
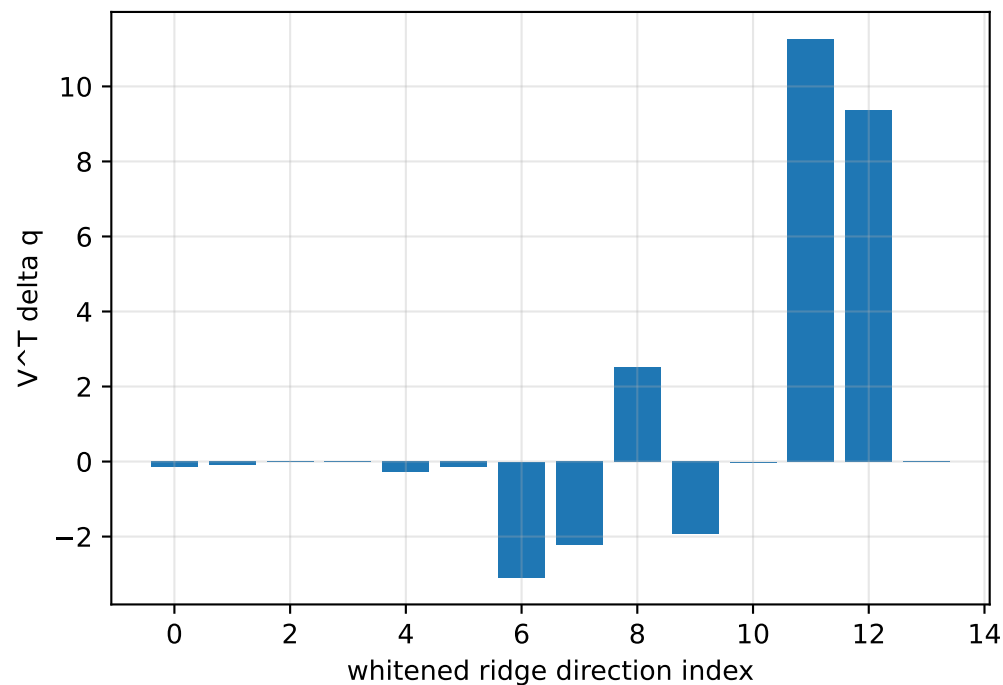
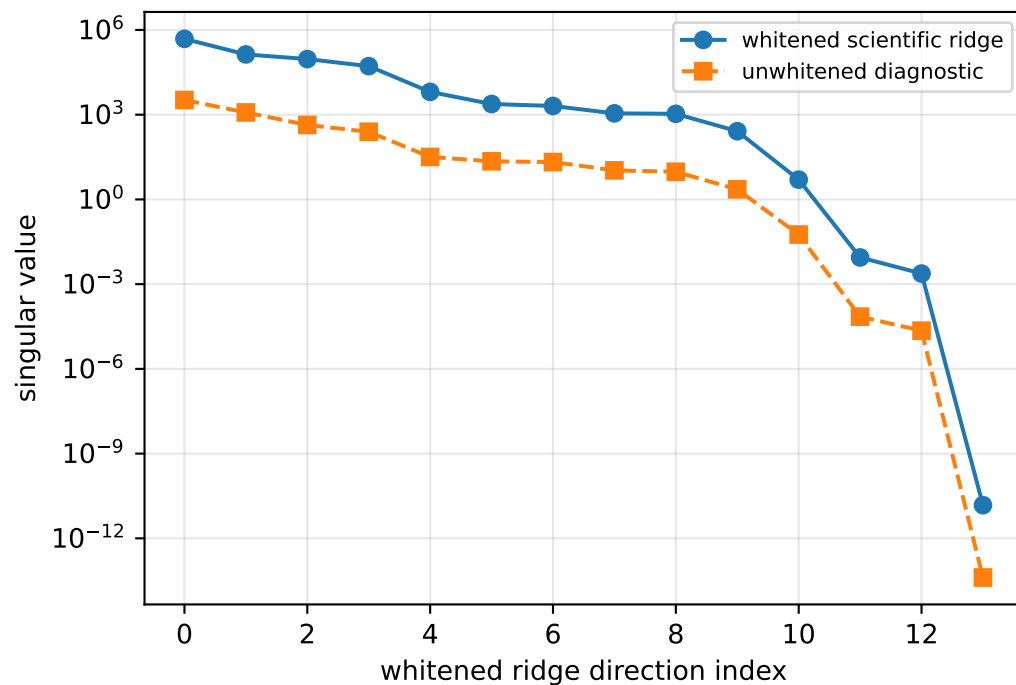
rotor lag [s] 0.2

gimbal lag [s] 0.049571991

sweep_sg_window_covariance_002: covariance weighting

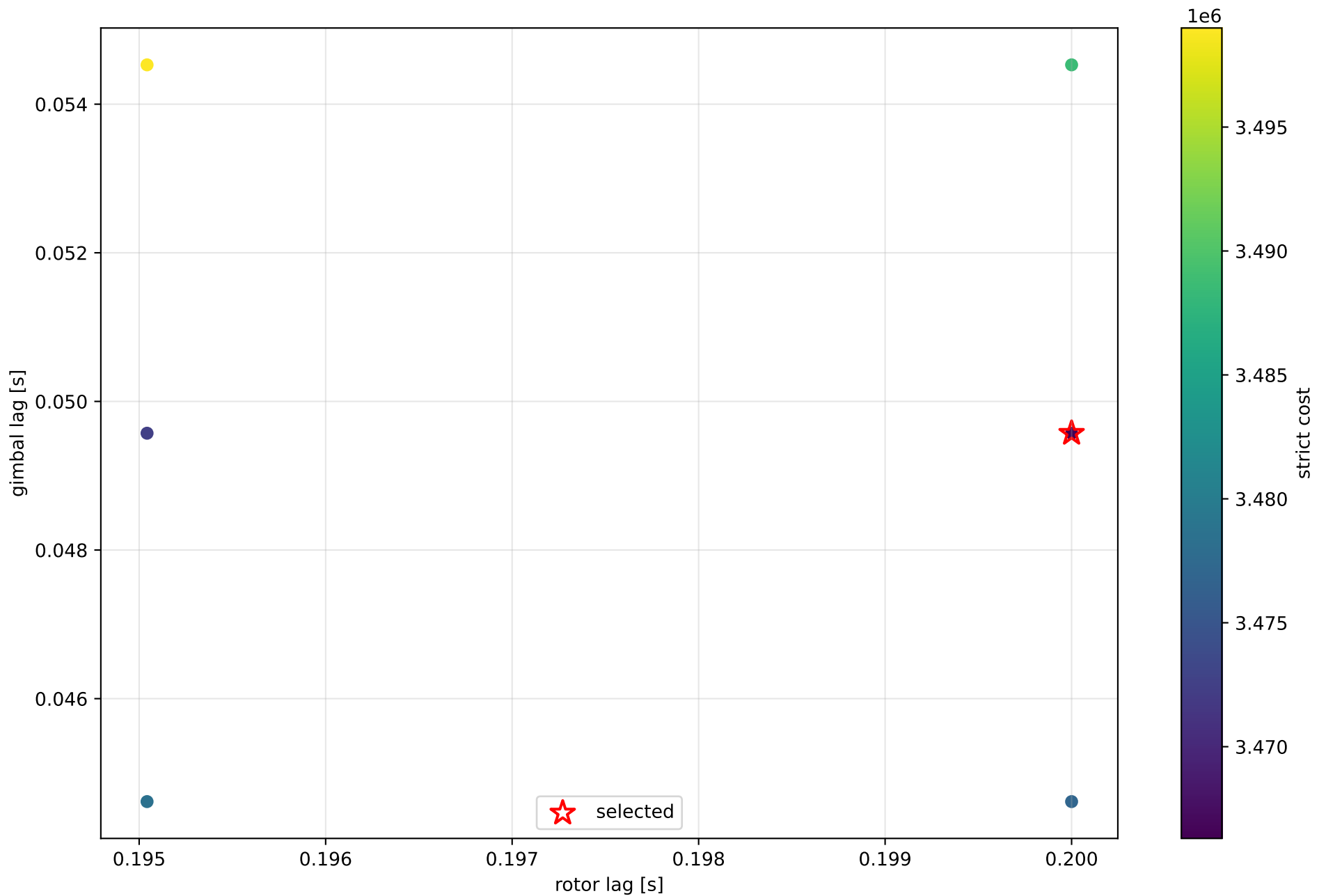


sweep_sg_window_covariance_002: parameter ridge



exact scale gauge: $||v_scale||=3.65778e-11$, rank=13, nullity=1

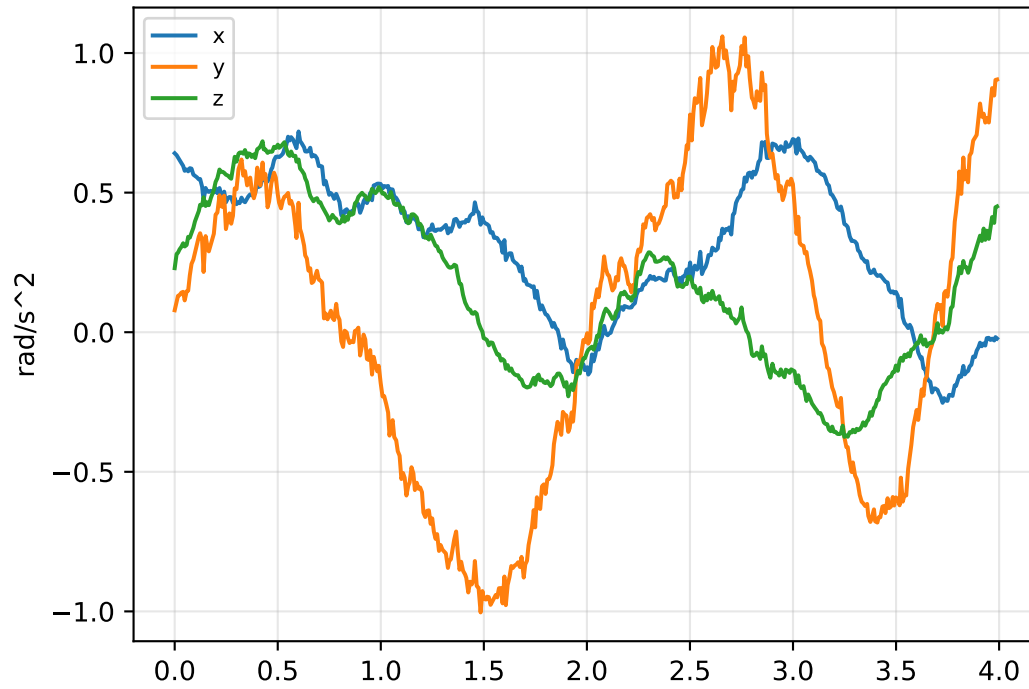
sweep_sg_window_covariance__002: strict lag candidates



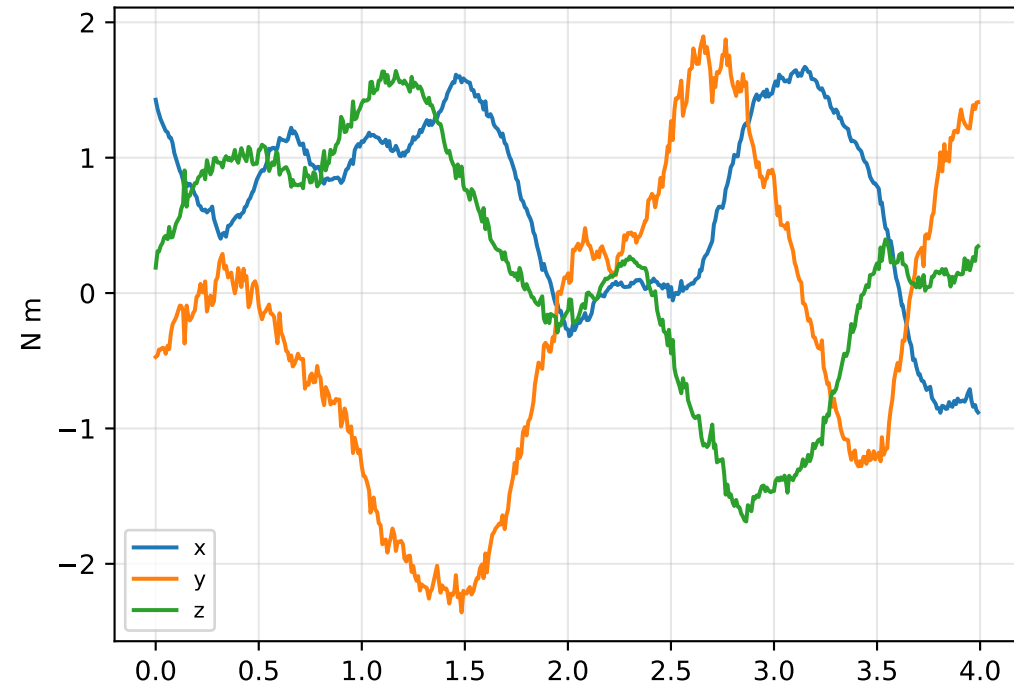
boundary flags: rotor lower=False upper=True; gimbal lower=False upper=False

sweep_sg_window_covariance_002: acceleration/wrench closure

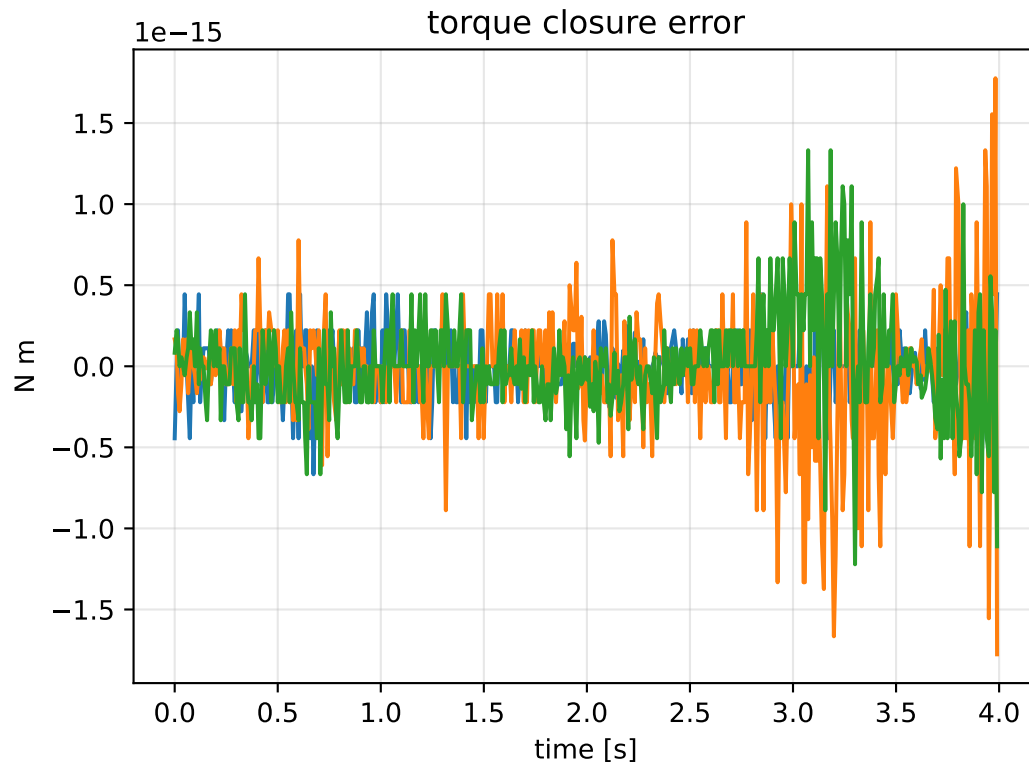
angular-acceleration residual



raw torque residual



torque closure error



```
principal moments [kg m^2]
nominal: [0.064953 0.065 0.128992]
estimated: [0.893267 2.934504 3.827759]
estimated / nominal: [13.752595 45.146171 29.674366]

force closure max/rms: 4.92939e-14 / 9.69573e-15
torque closure max/rms: 1.77636e-15 / 3.07422e-16
```